



From Compliance to Strategy: Cleaner Production as a Pathway to Sustainable Competitiveness



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Abstract: Cleaner production (CP) has evolved from a regulatory obligation into a strategic management approach that supports firms' transition toward sustainable competitiveness. This study examines how cleaner production practices move beyond compliance-oriented environmental management to become strategic capabilities associated with stronger innovation orientation, operational efficiency, and corporate legitimacy. Drawing on the Natural Resource-Based View (NRBV) and the Porter Hypothesis, this research employs a multiple-case qualitative content analysis of six energy-intensive firms—Enerjisa, Tüpraş, Şişecam, Enel, E.ON, and TotalEnergies—for the 2020–2024 reporting period. The findings identify a three-stage evolutionary trajectory of cleaner production integration: compliance-driven, efficiency-driven, and strategy-driven. Firms that move toward strategic integration tend to exhibit stronger dynamic and organisational capabilities and clearer strategic positioning in sustainability-oriented decision-making. European multinationals demonstrate more holistic and mature integration due to stable policy frameworks and access to sustainable finance, whereas emerging-economy firms primarily leverage cleaner production for efficiency gains and regulatory compliance. The study contributes to theory by conceptualising cleaner production as a dynamic strategic capability rather than a technical or operational tool and by providing comparative qualitative evidence consistent with the innovation-competitiveness mechanisms proposed by the Porter Hypothesis across diverse institutional contexts. In practice, the findings offer actionable insights for managers and policymakers seeking to design regulatory, financial, and organisational enablers that accelerate the transition from compliance to strategy, thereby positioning cleaner production as a central pathway through which firms may build sustainable competitiveness. Given the qualitative and document-based design, the study does not claim causal proof but identifies patterned associations across cases and institutional contexts.

Keywords: Cleaner production; Sustainable competitiveness; Natural Resource-Based View; Strategic management

1. Introduction

Cleaner production (CP) has evolved from a narrowly defined environmental management tool into a central pillar of sustainability-oriented corporate transformation. Initially promoted by international organisations as a preventive approach to reducing waste and emissions, CP was widely implemented as a technical and regulatory requirement aimed at minimising environmental impact at the lowest possible cost (Hens et al., 2018; Leeraphong et al., 2025; Severo et al., 2017). During this early compliance-oriented period, firms often adopted end-of-pipe technologies and incremental efficiency measures that remained peripheral to strategic decision-making. As regulatory pressures increased and stakeholder expectations evolved, CP gained renewed importance as a framework linking environmental quality with operational and organisational improvements.

Parallel developments in strategic management theory have reframed the relationship between environmental practices and competitiveness. The Resource-Based View (RBV) and its extension, the Natural Resource-Based View (NRBV), argue that pollution prevention, resource efficiency, product stewardship, and sustainability-oriented capabilities can generate resources that are valuable, rare, and inimitable, thereby underpinning long-term competitive advantage (Barney et al., 2011; Hart, 1995; Russo & Fouts, 1997). Similarly, Porter & van der Linde

(1995) proposed that environmental regulation, when properly designed, can stimulate innovation, productivity, and profitability—challenging traditional assumptions about the trade-off between environmental and economic performance. These perspectives collectively position environmental practices not as compliance burdens, but as strategic enablers capable of enhancing a firm's adaptability, resilience, and capacity for innovation.

Recent empirical research has reinforced these theoretical claims by demonstrating that CP practices—ranging from resource efficiency and waste minimisation to circular economy initiatives and process redesign—are reported to be associated with operational, environmental, and economic benefits (Al Kaabi et al., 2025). Firms integrating CP into their management systems, quality structures, and innovation processes are more likely to achieve eco-efficiency, respond effectively to market and regulatory changes, and strengthen their competitive position (Oliveira Neto et al., 2020). Despite these advancements, CP is often treated as a technical or operational tool rather than a strategic pathway for building sustainable competitiveness.

At the same time, sustainable competitiveness has emerged as a key framework in policy and management research. Rather than focusing solely on costs, productivity, or short-term profits, sustainable competitiveness emphasises the capacity to generate long-term economic value while safeguarding environmental integrity and social well-being (Balkytė & Tvaronavičienė, 2010). Recent studies highlight innovation, environmental, social, and governance (ESG) performance, resilience, and sustainability-oriented capabilities as foundational elements of sustainable competitiveness. However, CP is often included as a secondary indicator within composite indices rather than being conceptualised as an active, dynamic process that may contribute to sustainable competitiveness (Zhang et al., 2025).

Research on eco-innovation similarly confirms the potential of environmental innovations—new products, processes, digital tools, and circular business models—to improve firm competitiveness, particularly when integrated into strategic and organisational systems (Adams et al., 2016; Barriga Medina et al., 2022; Lewandowska et al., 2022). However, most eco-innovation studies emphasise innovation outputs (e.g., patents, Research and Development (R&D) intensity) rather than underlying CP pathways, organisational learning processes, and capability development mechanisms through which firms may shift from reactive compliance to more proactive strategic engagement and leadership (Cai & Li, 2018; Tariq et al., 2025).

These issues are especially salient in emerging and transition economies. Firms in such contexts often face fragmented regulatory environments, resource constraints, and limited access to green finance, which push them toward short-term compliance rather than long-term strategic investment (Huang, 2025). At the same time, growing exposure to global supply chains, ESG requirements, and low-carbon market opportunities creates pressure for deeper organisational change and leadership in CP (Mohammed, 2026). Understanding how companies across differing institutional contexts navigate the transition from compliance-driven CP to strategy-driven CP constitutes a notable theoretical gap.

Despite substantial research on CP, eco-innovation, environmental management systems, and sustainable competitiveness, several critical gaps remain. First, existing CP literature tends to emphasise technical options, operational efficiency, and environmental performance, rather than treating CP as a strategic capability that enhances long-term competitiveness (Mady et al., 2023). Second, limited empirical evidence documents how CP both evolves along a compliance-efficiency-strategy continuum and aligns with capability development, as outlined in NRBV theory (Alkaraan et al., 2024; Hart, 1995). Third, little comparative research has investigated how institutional environments shape the development of CP practices in emerging and developed economies, thereby limiting the understanding of the conditions under which CP contributes most effectively to sustainable competitiveness (Geng et al., 2025).

Against this backdrop, this study examines how CP evolves from a compliance-oriented environmental practice into a strategically embedded organisational capability. Drawing on the NRBV and the Porter Hypothesis, the study develops and empirically explores a compliance-efficiency-strategy trajectory, using qualitative content analysis of six energy-intensive firms operating in different institutional contexts.

The study does not seek statistical generalisation. Instead, it aims for analytical generalisation by tracing how CP is framed, implemented, and linked to selected proxies of sustainable competitiveness across theoretically relevant cases. These proxies include reported operational-efficiency gains, ESG-related positioning, access to sustainable finance, integration with innovation and circularity initiatives, and evidence of strategic embedding in governance and investment decisions.

This study addresses these gaps by examining how CP evolves from a compliance-oriented environmental practice into a strategic capability that may support sustainable competitiveness. Grounded in the NRBV and informed by the literature on CP, eco-innovation, and sustainability-oriented capabilities, this study analyses how firms integrate CP into their strategic processes, which organisational mechanisms support this integration, and how these mechanisms are associated with economic, environmental, and reputational outcomes. The central research question guiding this paper is: How can CP be transformed from a compliance-oriented environmental practice into a strategic approach that enhances firms' sustainable competitiveness?

This paper makes three main contributions. First, it conceptualises CP as a dynamic strategic capability, extending the NRBV and the sustainability-oriented innovation theory. Second, it develops and empirically

illustrates a compliance-efficiency-strategy trajectory for CP, identifying distinct practices, capabilities, and performance outcomes throughout this evolution. Third, it offers practical and policy insights for firms—particularly in emerging markets—seeking to leverage CP not only to meet regulatory requirements but also to strengthen their long-term, sustainability-driven competitive position.

2. Literature Review

CP emerged in the early 1990s as a preventive, efficiency-oriented strategy designed to reduce waste, emissions, and resource use throughout production systems. Unlike end-of-pipe approaches, CP integrates environmental considerations directly into product and process design, enabling firms to achieve ecological benefits alongside cost and productivity gains. Early empirical evidence demonstrated that CP measures—such as energy recovery, waste minimisation, and material substitution—generated measurable operational efficiencies, illustrating that environmental responsibility and profitability can complement one another (Salami & Klinghoffer, 2025). However, much of this literature still treats CP primarily as a preventive or efficiency-oriented production practice rather than as a staged strategic capability-development process.

As CP practices matured, the concept expanded beyond technical interventions to encompass broader management principles that support sustainability-oriented innovation, organisational learning, and circular resource flows (Ghormare et al., 2024; Nasrollahi et al., 2020). Contemporary CP research increasingly links CP with eco-innovation, environmental management systems, life-cycle thinking, and circular economy practices. Yet these studies rarely explain how such practices become embedded in strategic decision-making over time. Adoption drivers range from regulatory pressures and stakeholder expectations to technological innovation and firm capabilities (Maldonado et al., 2025; Tu et al., 2025). In emerging economies, CP is increasingly recognised as a means of overcoming institutional constraints and improving competitiveness (Nasrollahi et al., 2020; Wang et al., 2023).

The transition from compliance to strategy represents a major paradigm shift in corporate environmental management. Earlier work has shown that firms often view environmental practices as necessary costs incurred to satisfy regulatory requirements (Haile & Singh, 2025). However, seminal contributions by Hart (1995) and Porter & van der Linde (1995) reframed environmental practices as potential drivers of innovation and competitiveness. Subsequent studies have demonstrated how integrating environmental management into strategic planning leads to superior financial, reputational, and operational outcomes (Barney et al., 2011; Russo & Fouts, 1997). CP, as an operational embodiment of this integration, enables firms to build dynamic capabilities associated with continuous innovation, stakeholder engagement, and resource efficiency (Andersén, 2021). Empirical studies suggest that proactive environmental strategies are associated with efficiency gains, reputational benefits, and improved risk management. However, the specific pathway through which CP moves from compliance to strategic integration remains underexplained.

Research on eco-innovation reinforces the strategic potential of CP by demonstrating that environmental innovations—such as new green products, resource-efficient processes, and circular business models—strengthen both environmental and economic performance (Adams et al., 2016; De Marchi, 2012). Eco-innovation studies generally associate environmental innovation with stronger market positioning, stakeholder legitimacy, and sustainability-related performance (Méndez-León et al., 2024; Shahbaz & Malik, 2025). Nevertheless, this literature often focuses on innovation outputs, such as green products, patents, or R&D activity, rather than on the CP routines and organisational learning processes that precede strategic integration.

CP provides the operational foundation for eco-innovation, while eco-innovation adds technological and strategic depth; together, they form the core of sustainability-oriented innovation (Alkaraan et al., 2024; Nasrollahi et al., 2020; Severo et al., 2017). However, the effectiveness of this relationship depends on internal capabilities and external institutional conditions (Rifai et al., 2025). In emerging economies, institutional voids and resource constraints often hinder sustained investment in CP and eco-innovation, leading to uneven adoption and performance variability (Austin et al., 2025). Therefore, the CP–eco-innovation link needs to be examined not only as a performance relationship, but also as a capability-building process shaped by organisational and institutional conditions.

Sustainable competitiveness extends traditional competitiveness by integrating economic performance, environmental stewardship, and social resilience (Balkytė & Tvaronavičienė, 2010; Wang et al., 2023). At the firm level, it refers to the ability to generate long-term value while managing resource constraints, stakeholder pressures, and systemic risks (Cavaco & Machado, 2015). Innovation, digitalisation, and ESG-oriented governance systems have emerged as critical determinants of sustainable competitiveness (Marczewska et al., 2025).

CP contributes directly to sustainable competitiveness by enhancing resource efficiency, embedding sustainability within organisational routines, and strengthening corporate legitimacy. Firms with integrated CP practices and strong sustainability-oriented capabilities tend to outperform their peers in productivity, innovation, and market competitiveness (Larabi, 2026; Pereira-Moliner et al., 2012). Sustainable competitiveness can thus be understood as an emergent outcome of long-term CP integration coupled with strategic intent and organisational

learning.

Despite the growing literature on CP, eco-innovation, and sustainable competitiveness, these research streams remain only partially integrated. CP studies often emphasise technical options, operational efficiency, and environmental performance, while eco-innovation research tends to focus on innovation outputs rather than the underlying organisational routines that enable strategic transformation. Similarly, sustainable competitiveness studies frequently treat environmental practices as aggregate indicators instead of examining how they become embedded in firm-level strategy. As a result, limited research explains how CP evolves from compliance-oriented adoption into a strategically embedded capability, particularly across different institutional contexts. To address this gap, this study conceptualises CP as a staged strategic capability-development process. Building on the NRBV and sustainability-oriented innovation literature, it develops a compliance–efficiency–strategy trajectory and examines how this trajectory appears across six energy-intensive firms operating in different institutional contexts.

3. Methodology

3.1 Research Design

This study employs a qualitative, multiple-case, content-analysis design to investigate how CP evolves from a compliance-oriented practice into a strategic capability that supports sustainable competitiveness. A qualitative approach enables an in-depth examination of organisational learning processes, managerial interpretations, and contextual dynamics, while a multiple-case design ensures analytical generalisation across diverse regulatory and industry settings (Krippendorff, 2018; Mayring, 2021; Wójcik et al., 2025). The research is anchored in the NRBV (Barney et al., 2011; Hart, 1995) and the sustainability-oriented innovation literature (Adams et al., 2016), which jointly inform the analytical framework and the interpretation of the findings.

3.2 Case Selection and Scope

A purposive sampling strategy was adopted to ensure heterogeneity and cross-contextual validity. Six firms operating in energy-intensive and manufacturing sectors were selected based on three criteria: (a) public availability of detailed sustainability and cleaner-production disclosures; (b) operations across distinct institutional and regulatory environments; and (c) availability of sustainability and integrated reports covering the 2020–2024 reporting period.

The final sample comprises three European multinationals—Enel, E.ON, and TotalEnergies—and three Turkish firms—Enerjisa, Tüpraş, and Şişecam. Data were collected from corporate sustainability reports, integrated reports, policy documents, ESG disclosures, and third-party assessments (CDP, Sustainalytics, MSCI, and Refinitiv). This longitudinal dataset provides robust evidence of changes in CP practices and strategic integration over a five-year period.

This study deliberately focuses on large, publicly listed, energy-intensive firms rather than small- and medium-sized enterprises (SMEs). This sampling decision is motivated by theoretical and methodological considerations.

First, large firms operating in energy-intensive sectors are more likely to implement CP practices across multiple organisational layers, ranging from regulatory compliance and operational efficiency to strategic integration. As such, they provide a suitable empirical context for observing the full compliance-efficiency-strategy trajectory proposed. SMEs, by contrast, often adopt CP in a fragmented or project-based manner, which limits visibility into long-term capability development.

Second, publicly listed firms are subject to stringent disclosure requirements and regularly publish sustainability, integrated, and ESG reports aligned with international standards such as the Global Reporting Initiative (GRI). This ensures data availability, transparency, and longitudinal comparability—key requirements for qualitative content analysis and cross-case synthesis.

Third, the strategic relevance of CP is more pronounced in large firms due to their capital intensity, regulatory exposure, and stakeholder scrutiny. These firms face greater pressure from investors, regulators, and global value chains, which makes them more likely to translate CP from an operational necessity to a strategic capability.

Accordingly, the selected cases are not intended to represent all firm types statistically, but rather to enable analytical generalisation by capturing theoretically rich and information-dense cases. This approach aligns with established methodological guidance for multiple-case qualitative research, where depth, comparability, and theoretical relevance take precedence over population representativeness.

These six firms were selected not merely because they publish sustainability reports, but because they provide theoretically relevant variation across three dimensions: sectoral energy intensity, visibility of CP-related disclosures over time, and institutional contrast between mature EU sustainability regimes and an emerging-economy setting. Within the Turkish context, Enerjisa was selected because it offers clear evidence of electricity-distribution dynamics and digital-efficiency initiatives, Tüpraş because it represents a highly energy- and emissions-intensive refining context, and Şişecam because it provides a relevant manufacturing case characterised

by circularity practices and fuel-substitution efforts. Within the European context, Enel was selected for its strong emphasis on circular strategy and strategic embedding, E.ON for its system-level integration of network transformation and digital infrastructure, and TotalEnergies for its transition-oriented reinvestment logic and low-carbon repositioning. Taken together, these cases provide a comparative basis for tracing how CP is framed, implemented, and strategically embedded across different sectors and institutional environments.

3.3 Data Collection and Coding Procedures

The study employs a hybrid content analysis approach, systematically combining deductive and inductive coding procedures (Krippendorff, 2018; Mayring, 2021). The deductive framework was derived from two streams of the literature: the NRBV (Barney et al., 2011; Hart, 1995) and the established CP dimensions. Three overarching categories were defined:

- (1) Regulatory Compliance—legal adherence, environmental reporting, ISO 14001 certification.
- (2) Operational Efficiency—waste reduction, resource optimisation, cost-saving process improvements.
- (3) Strategic Integration—innovation initiatives, stakeholder engagement, ESG-linked investments.

To ensure cross-case comparability, these categories were mapped onto the relevant GRI Standards (301–308; 401–419).

Inductive coding involved iterative readings, resulting in subcodes, including circular product design, digitalised production, green R&D trajectories, and sustainability-oriented supply-chain collaboration. Two independent coders applied all codes using NVivo 14; intercoder reliability reached Cohen's $\kappa = 0.87$, indicating high agreement. Discrepancies were resolved through consensus, ensuring the validity of category definitions and coding logic.

To improve classification transparency, Table 1 expands the coding structure by explicitly linking first-order codes and second-order categories to stage-assignment rules, dynamic capability indicators, and selected proxies of sustainable competitiveness. This structure ensures that case classification is based not on isolated disclosures, but on repeated, theory-informed patterns observed across documents and reporting periods.

Table 1. Overview of the coding logic, stage assignment, and analytical structure

Analytical Level	Description	Example	Stage Assignment Rule	Dynamic Capability Indicator	Sustainable Competitiveness Proxy
Data source	Corporate sustainability, integrated, annual, and ESG reports published between 2020 and 2024, supported by selected third-party ESG assessments	Tüpraş Integrated Annual Report 2025; Enel Integrated Annual Report 2025; CDP/Sustainalytics summaries	Sources were included only when they provided repeated and traceable CP-related disclosures across reporting periods	Repeated evidence of sustainability-related opportunity recognition, investment action, and organisational embedding	Reported efficiency gains, ESG positioning, green finance alignment, innovation integration, strategic sustainability positioning
First-order codes	Descriptive codes capturing explicit Cleaner Production (CP)-related statements in corporate disclosures	“ISO 14001 certification”; “energy intensity reduction”; “digital metering”; “cullet recycling”; “green bond issuance”	Individual codes were not used alone for final stage classification; they had to recur across documents and years	Sensing-related codes: risk/opportunity recognition; Seizing-related codes: investment/process redesign; Reconfiguring-related codes: governance, supply-chain, or strategy integration	Explicit references to operational savings, emissions/intensity reductions, ESG improvements, investor relevance, product/process innovation
Second-order categories	Aggregated categories reflecting underlying CP mechanisms across cases	Regulatory compliance; Operational eco-efficiency; Digital integration; Circular innovation; ESG-linked investment	Categories were used to identify whether CP remained compliance-oriented, became efficiency-oriented, or reached strategy-level integration	Pollution prevention; Resource efficiency; Organisational learning; Strategic reinvestment; Capability reconfiguration	Eco-efficiency outcomes, reputational signals, access to sustainable finance, low-carbon strategic repositioning
Analytical dimensions	Higher-order dimensions derived through axial coding and cross-case	Compliance-driven CP; Efficiency-driven CP; Strategy-driven CP	Compliance-driven: CP framed mainly as legal adherence, certification, and legitimacy.	Sensing: identifying sustainability pressures/opportunities. Seizing: implementing CP investments and redesign.	Evidence of stronger ESG orientation, innovation capacity, efficiency performance, green-

Theoretical alignment	comparison		Efficiency-driven: CP linked to cost savings, resource optimisation, and process improvement. Strategy-driven: CP linked to innovation, capital allocation, governance, and long-term value creation	Reconfiguring: embedding CP into governance, innovation, and business strategy	finance relevance, and long-term strategic embedding
	Linkage of empirical categories to the NRBV, dynamic capabilities perspective, and Porter Hypothesis	Pollution prevention; Dynamic capabilities; Innovation-led competitiveness	Final interpretation required alignment between coded disclosures and theory-informed stage criteria rather than author judgement alone	NRBV: pollution prevention and strategic environmental capability; Dynamic capabilities: sensing, seizing, reconfiguring	Sustainable competitiveness interpreted through observable qualitative proxies rather than direct causal measurement

Note: CP = Cleaner production; ESG = Environmental, social, and governance; NRBV = Natural Resource-Based View.

3.4 Data Analysis

Data analysis followed a three-stage, theory-driven sequence to ensure systematic rigor and cross-case comparability.

Stage 1: Descriptive Coding

All explicit and implicit CP practices were identified and categorised to construct within-case profiles of compliance, efficiency, and strategic actions.

Stage 2: Pattern Coding

Descriptive codes were condensed into higher-order themes linking CP activities to competitive outcomes-such as cost reduction, process and product innovation, and improvements in corporate reputation. These themes reflect NRBV constructs and sustainability-oriented innovation pathways.

Stage 3: Cross-Case Synthesis

Patterns were examined using constant-comparative techniques across developed (EU) and emerging (Türkiye) contexts to identify structural differences, contextual enablers, and capability trajectories. This synthesis informed the development of a three-phase compliance-efficiency-strategy model of the evolution of CP.

Triangulation of academic literature, corporate press releases, and third-party ESG ratings enhanced the analytical robustness and reduced single-source bias (Alkaraan et al., 2024; Arvidsson & Dumay, 2022). The resulting framework provides an empirically grounded explanation of how CP transforms from an operational requirement into a strategic capability.

Case-stage classification was not based on isolated disclosures but on repeated evidence across multiple reporting years and categories. A firm was coded as strategy-driven only when CP-related actions were explicitly linked to innovation, investment, governance, or long-term value creation, rather than to compliance or efficiency alone.

3.5 Validity, Reliability, and Ethical Considerations

Because sustainability reports may contain symbolic or impression-management disclosures, the analysis did not treat corporate statements as self-validating evidence. Wherever possible, claims were cross-checked against third-party ESG assessments, repeated disclosures across years, and observable strategic signals such as green bond activity, taxonomy alignment, or sustained capital allocation toward low-carbon initiatives.

Construct validity was ensured through multi-source triangulation using corporate documents, ESG assessments, and peer-reviewed research. External validity was strengthened by benchmarking the findings against established frameworks, including Porter & van der Linde (1995)'s innovation hypothesis and the World Economic Forum's sustainable competitiveness pillars (Balkytė & Tvaronavičienė, 2010). Reliability was enhanced through an audit trail documenting coding decisions, NVivo structures, and version-controlled memos. Because the study relied exclusively on publicly available documents, it did not involve human subjects or confidential data, thereby ensuring full compliance with ethical standards. Potential reporting biases were mitigated by triangulation with third-party ESG evaluations. Nevertheless, the document-based design cannot fully eliminate greenwashing risk, which should be considered when interpreting stage maturity and competitiveness outcomes.

3.6 Methodological Contribution

Methodologically, the study advances CP research by applying a longitudinal multiple-case content analysis that captures both the temporal evolution and the institutional variation in CP practices. The integration of NRBV with sustainable competitiveness frameworks bridges environmental management research and strategic capability development. By operationalising the compliance-efficiency-strategy trajectory using GRI-aligned indicators and transparent coding procedures, the study offers a replicable analytical model to examine the maturation of CP capabilities. The methodological rigour-ensured through triangulation, explicit coding structures, and a detailed audit trail-strengthens the credibility of the findings and demonstrates how CP can evolve into a strategic capability that supports sustainability outcomes.

4. Findings

To enhance analytical transparency and address potential subjectivity in interpretation, Table 2 presents a structured cross-case comparison of CP practices, coding categories, stage assignment logic, dynamic capability indicators, and associated competitiveness proxies.

The findings reveal that CP adoption does not follow a uniform trajectory across firms, but instead reflects differentiated patterns of capability development. Turkish firms (Tüpraş, Şişecam, Enerjisa) predominantly exhibit compliance-driven and efficiency-driven CP characteristics, whereas European firms (Enel, E.ON, TotalEnergies) show stronger evidence of more advanced forms of CP integration, including elements of strategic embedding. However, this distinction should not be interpreted as a strict dichotomy, but rather as a continuum with intermediate positions.

Across all cases, stage classification is based not on isolated disclosures but on repeated patterns observed across reporting periods. A firm is classified as strategy-driven only when CP is explicitly embedded in innovation, governance, and long-term strategic positioning, rather than being limited to compliance or efficiency improvements.

Importantly, the relationship between CP maturity and sustainable competitiveness should be interpreted with caution. While the evidence suggests that more advanced CP integration is associated with stronger ESG positioning, innovation orientation, and efficiency improvements, the analysis relies on qualitative proxies rather than direct causal measurement. Therefore, the findings should be understood as indicative of a structured relationship rather than as definitive proof of causality.

Table 2. Cross-case evidence for cleaner production (CP) stage assignment and competitiveness proxies

Firm	Key CP Practice	Primary Disclosure Basis	Coding Category	Stage Assignment Rationale	Dynamic Capability Marker	Competitiveness Proxy	Caution/Limitation
Enerjisa	Digital metering, smart-grid investments, loss reduction, energy-flow optimisation	Annual report and sustainability -related disclosures referring to digital optimisation and network efficiency	Operational eco-efficiency; Digital integration	Enerjisa shows repeated evidence of CP being used beyond compliance and linked to efficiency and operational improvement. However, evidence of full organisation-wide strategic embedding is still partial rather than complete.	Sensing: identification of system inefficiencies and energy losses; Seizing: digital metering and smart-grid investments; Reconfiguring: partial integration into broader sustainability and network-management logic	Efficiency improvement, ESG-maturity signals, stronger sustainability positioning in energy distribution	Evidence supports a transition from compliance to efficiency, but full strategy-driven integration remains developing rather than fully established
				Tüpraş remains primarily positioned between compliance-driven and efficiency-driven CP. CP is repeatedly framed as necessary for operational legitimacy and process optimisation,	Sensing: recognition of environmental compliance requirements; Seizing: efficiency-oriented process improvements; Reconfiguring: limited evidence beyond operational	Process-efficiency gains, environmental legitimacy, resource optimisation	Strategic capability claims should be interpreted cautiously because disclosures more strongly
Tüpraş	ISO 14001-based environmental management, pollution control, heat-recovery systems, closed-loop water practices	Integrated annual report emphasising compliance systems, environmental management, and selected process-	Regulatory compliance; Operational eco-efficiency				

	efficiency improvement s		but evidence of strong innovation- or governance-level strategic embedding is limited.	adaptation		support compliance deepening and efficiency than full strategic transformation
Şişecam	Cullet recycling, alternative fuels, energy-intensity reduction, lower-carbon production practices	Sustainability and annual reporting highlighting recycling, fuel substitution, and emissions-intensity improvement	Operational eco-efficiency; Circular production practice	Şişecam demonstrates stronger efficiency-driven CP and some early strategic signals through circularity-related practices. However, evidence of CP being fully embedded in capital allocation or corporate-wide strategic governance remains limited.	Sensing: recognition of emissions and material-efficiency pressures; Seizing: recycling and alternative-fuel implementation; Reconfiguring: partial movement toward circular production routines	Resource circularity, lower-carbon production positioning, operational eco-efficiency
Enel	Circular economy integration, sustainability-oriented infrastructure planning, product design alignment, investment-related embedding	Integrated annual report explicitly linking circularity and sustainability to planning, design, and value creation	Strategic integration; Circular innovation; ESG-linked strategic embedding	Enel provides repeated and explicit evidence that CP-related practices are embedded in innovation, infrastructure planning, and long-term value creation. This justifies classification as strategy-driven CP.	Sensing: identification of sustainability opportunities and transition pressures; Seizing: integration of circular principles into operations and planning; Reconfiguring: embedding into design, investment, and governance processes	Strategic sustainability positioning, innovation orientation, stronger long-term value-creation framing, investor-facing ESG relevance
E.ON	Smart-grid alignment, intelligent distribution networks, renewable integration through digital infrastructure	Integrated annual report and sustainability disclosures linking network transformation to efficiency and decarbonisation	Digital integration; System-level efficiency; Emerging strategic embedding	E.ON shows advanced CP-related integration at the system and infrastructure level. Compared with Enel, however, the disclosures frame CP more as a platform for efficient system management than as a fully articulated corporate-wide CP strategy.	Sensing: identification of energy-system complexity and transition needs; Seizing: infrastructure modernisation and digital grid investment; Reconfiguring: integration of CP-related logic into intelligent network systems	Investor-facing ESG alignment, system-level operational efficiency, low-carbon infrastructure positioning
TotalEnergies	Reinvestment of efficiency gains into low-carbon transition, renewable expansion, CP linked to transformation strategy	Sustainability and climate reporting connecting efficiency, transition investments, and low-carbon strategic	Strategic integration; Capability reconfiguration; ESG-linked investment	TotalEnergies shows repeated evidence that CP-related efficiency gains are linked to broader transition strategy, investment redirection, and long-term repositioning. This supports	Sensing: identification of transition and decarbonisation pressures; Seizing: low-carbon and renewable-oriented investments; Reconfiguring: strategic redirection	Low-carbon strategic repositioning, investor relevance, transition-oriented innovation and reinvestment logic

direction	classification as strategy-driven CP.	of resources and capability base	reported transition outcomes are not fully comparable across all firms
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Reported efficiency- or cost-related figures are treated as firm-specific disclosure indicators rather than directly comparable performance measures, because firms differ in sector, reporting boundaries, baseline years, and calculation methods. The cross-case analysis suggests that institutional environments may be associated with differences in CP trajectories and strategic embedding.

European firms operate within stable regulatory frameworks, stringent disclosure obligations, and strong market incentives for decarbonisation. This institutional support appears to be associated with earlier and more cohesive integration of CP into strategy, innovation, and investment systems.

By contrast, Turkish firms face regulatory inconsistencies, limited enforcement, and constrained access to sustainable finance. Consistent with the emerging-economy literature, these conditions appear to be associated with stronger compliance- and efficiency-oriented practices, but slower progress in the strategic embedding of those practices.

Table 3 summarises the relative positioning of the six firms along the CP trajectory based on cross-case evidence. Stage assignment is not derived from isolated disclosures, but from repeated patterns observed across reporting periods and coding categories. The final column presents qualitative proxies of sustainable competitiveness, reflecting observable signals such as efficiency improvements, circularity practices, ESG-related positioning, and strategic embedding in investment and governance processes.

Table 3. Comparative positioning of firms along the cleaner production (CP) trajectory

Firm	Compliance-Driven	Efficiency-Driven	Strategy-Driven	Sustainable-Competitiveness Proxy
Enerjisa	✓	✓	△	Digital efficiency integration; ESG maturity signals
Tüpraş	✓	✓	△	Process efficiency and compliance deepening
Şişecam	✓	✓	△	Resource circularity and emissions-intensity reduction
Enel	—	✓	✓	Capital allocation and circular-strategy embedding
E.ON	—	✓	✓	System-level digital integration and investor-facing ESG alignment
TotalEnergies	—	✓	✓	Transition-oriented reinvestment and low-carbon strategic repositioning

Note: ✓ = strong implementation; △ = developing; — = limited.

Importantly, the classification should be interpreted as indicative rather than absolute. In several cases, firms exhibit partial or transitional characteristics (△), suggesting uneven progression across stages rather than a strictly linear trajectory.

This structured comparison enhances transparency by making the link between empirical evidence, coding logic, and theoretical interpretation explicit.

To illustrate the analytical categories identified in the cross-case analysis, this section presents selected excerpts from corporate sustainability and integrated reports. These excerpts provide illustrative evidence of how CP practices are framed, implemented, and progressively integrated across firms and institutional contexts.

In the compliance-driven stage, CP is predominantly framed in terms of regulatory alignment and certification. For instance, Tüpraş emphasises that environmental management is grounded in compliance-oriented systems, stating that “environmental compliance and adherence to national and international regulations remain the foundation of our sustainability approach, supported by ISO 14001-certified environmental management systems” (Tüpraş, 2025). This framing reflects a rule-based logic in which CP is positioned primarily as a mechanism for legitimacy and risk mitigation rather than as a strategic resource.

As firms progress toward efficiency-driven integration, CP practices are more frequently framed in relation to operational improvements and reported performance outcomes. Şişecam reports that “cullet recycling and alternative fuel use have reduced energy intensity and carbon emissions across selected production lines” (Şişecam, 2025), highlighting how CP contributes to eco-efficiency and cost reduction through process optimisation. Similarly, Enerjisa notes that “digital metering and smart-grid investments enable real-time optimisation and

reported loss-reduction” (Enerjisa, 2025), illustrating how digitalisation supports the routinisation of CP within operational systems.

European firms exhibit more advanced CP integration at the systems and infrastructure levels. EON (2025) explicitly frames its distribution networks as enablers of cleaner and more efficient energy systems, stating that “distribution networks are being transformed into intelligent platforms that manage complex energy and data flows, enabling efficient integration of renewable energy”. This indicates an advanced, efficiency-driven model in which CP is embedded within network management and digital infrastructure rather than being articulated as a standalone environmental programme.

At the strategy-driven stage, CP is explicitly linked to corporate strategy, capital allocation, and long-term value creation. Enel highlights that “circular economy principles are embedded in infrastructure planning, product design, and investment decisions, positioning sustainability as a core driver of value creation” (Enel, 2025), thereby suggesting a more advanced level of CP into corporate decision-making processes. Likewise, TotalEnergies frames CP as a strategic lever for transformation, noting that “reported efficiency improvements from CP initiatives are reinvested in low-carbon energy projects and innovation pathways” (TotalEnergies, 2025), thereby linking operational efficiency to capability reconfiguration and long-term competitiveness.

Taken together, these excerpts provide illustrative support for the compliance-efficiency-strategy trajectory identified in the cross-case analysis. They illustrate how CP may evolve from a compliance mechanism into an operational capability and, in more advanced cases, into a strategically embedded resource. Three recurring mechanisms appear to support firms’ progression along the CP trajectory. First, standardisation through ISO and GRI frameworks may create a foundation for more adaptive ESG and innovation-oriented management systems. Second, reported efficiency gains and process improvements are sometimes linked in corporate disclosures to green R&D, digitalisation, and sustainable product development. Third, CP routines appear to become embedded in training, supplier collaboration, and cross-functional sustainability practices, thereby supporting organisational learning. Three overarching insights summarise the empirical findings:

(1) CP follows an evolutionary pathway, shifting from regulatory compliance to internal efficiency and ultimately to strategic integration.

(2) CP appears to be associated with dual value signals by simultaneously enhancing internal operations and strengthening external legitimacy through improved innovation and ESG performance.

(3) Institutional context strongly influences CP outcomes, with firms in supportive policy and financial ecosystems achieving more advanced strategic integration.

Overall, the evidence suggests that CP is not merely a technical or environmental function but a strategic management process. Firms that embed CP within innovation systems, governance structures, and stakeholder relations appear more likely to report stronger operational, environmental, and reputational signals.

5. Discussion

This section interprets the empirical findings through the theoretical lenses of the NRBV, the Porter Hypothesis, and the sustainable competitiveness literature. The results suggest that CP does not operate as a static environmental practice but evolves into a strategic capability that may support firms’ competitiveness across diverse institutional contexts.

5.1 From Compliance to Strategic Integration

The findings suggest that CP develops through an evolutionary capability-building trajectory. In the initial stage, firms tend to adopt CP primarily in response to environmental regulations, certification requirements, and external enforcement pressures. This compliance-driven orientation, which is particularly visible among the Turkish firms in the sample, is consistent with prior research identifying regulatory pressure as a key driver of CP adoption (Habib et al., 2022; Li, 2025; Oliveira Neto et al., 2020; Tatoglu et al., 2019).

As CP becomes more deeply embedded in organisational routines, firms begin to internalise environmental practices in ways that generate process efficiency, operational learning, and accumulated tacit knowledge. These internal gains create the conditions for a broader strategic role of CP. In this sense, the progression from compliance to strategic integration reflects Hart (1995)’s NRBV, which proposes that environmental practices can evolve into valuable and difficult-to-imitate organisational capabilities when linked to core business activities. Rather than remaining a short-term regulatory response, CP increasingly may become part of a longer-term capability-development process (Andersén, 2021; Barney et al., 2011).

5.2 Cleaner Production as a Dynamic Capability

The findings indicate that CP may function as a dynamic capability when it supports resource reconfiguration, adaptive responsiveness, and continued innovation in the face of sustainability-related pressures. This pattern is

particularly visible in firms such as Enel, E.ON, and TotalEnergies, where CP is connected to digital transformation, R&D programmes, human capital development, and sustainability governance structures (E.ON, 2025; TotalEnergies, 2025). Such alignments appear to enhance absorptive capacity and strengthen firms' ability to sense and seize emerging sustainability opportunities (Alkaraan et al., 2024; Russo & Fouts, 1997).

From this perspective, CP is not simply an environmental tool but an evolving set of organisational routines that may be associated with environmental, operational, reputational, and financial value over time. This interpretation remains consistent with the NRBV, while adding a stronger process-oriented understanding of how CP becomes strategically meaningful as it is embedded more deeply within organisational systems. These findings are also in line with prior research showing that environmental capabilities contribute more effectively to firm performance when supported by managerial commitment and complementary assets (Andersén, 2021; Aryee et al., 2025; Yi & Demirel, 2023).

5.3 Qualitative Evidence Consistent with the Porter Hypothesis

The findings provide qualitative evidence broadly consistent with the Porter & van der Linde (1995) hypothesis. Across the cases, CP-related initiatives were frequently framed as cost-saving and efficiency-enhancing, although directly comparable firm-level financial effects were not uniformly disclosed.

Strategic uses of CP, such as Enel's circular economy initiatives and TotalEnergies' renewable energy expansion, are consistent with Porter's "innovation compensation effect". Rather than imposing competitive burdens, environmental expectations appear to be associated with technological and organisational creativity. This pattern is consistent with contemporary findings indicating that proactive sustainability strategies yield greater innovation and long-term competitive advantage (Čater et al., 2025; Hermundsdottir & Aspelund, 2022; Javeed et al., 2022; Villegas et al., 2026).

5.4 Cleaner Production and Sustainable Competitiveness

The findings suggest that more advanced CP integration is associated with stronger qualitative signals of sustainable competitiveness, defined as the ability to maintain a long-term competitive advantage while simultaneously improving environmental and social outcomes (Balkytė & Tvaronavičienė, 2010; Kusuma & Jumanoro, 2025). Firms that reached the strategy-driven stage reported stronger signals of innovation capacity, gained greater stakeholder legitimacy, achieved higher ESG scores, and strengthened their market positioning.

European firms benefited from coherent regulatory frameworks, predictable disclosure requirements, and access to sustainable finance, enabling them to integrate CP into strategic planning and investment processes. Meanwhile, Turkish firms exhibited strong efficiency-based gains but limited strategic embedding, reflecting weaker regulatory enforcement, funding constraints, and institutional gaps (Nipper et al., 2025). These contrasts affirm the importance of institutional maturity in accelerating the strategic contribution of CP.

5.5 Theoretical, Managerial, and Policy Implications

This study contributes to the academic literature in three key ways. First, it extends the NRBV by interpreting CP as an evolving capability whose strategic significance depends on organisational embedding and institutional context. Second, it integrates CP into the sustainable competitiveness framework, suggesting how environmental routines contribute to a long-term strategic advantage. Third, by providing evidence from both developed and emerging economies, the study offers contextual validation of the Porter Hypothesis, providing qualitative evidence that environmental practices may be associated with innovation and competitiveness even in the presence of institutional constraints.

For managers, the findings underscore the importance of treating CP as a strategic investment rather than a compliance obligation. Integrating CP with R&D, digital transformation, supply chain governance, and ESG reporting may support significant innovation potential, operational efficiencies, and reputational benefits. Strengthening ESG transparency may enhance investor confidence, support access to sustainable finance, and position firms more competitively in sustainability-driven markets.

At the policy level, the study highlights the need for stable and coherent regulatory frameworks that encourage firms to pursue strategic-rather than reactive-environmental action. Financial instruments such as tax incentives, green financing schemes, and technology transfer programs can accelerate firms' strategic adoption of CP. Additionally, strengthening ESG disclosure standards and benchmarking systems can enhance transparency and create competitive pressure that drives improved sustainability performance across industries.

Overall, the findings suggest that CP has evolved from a technical environmental tool into a transformative strategic capability. As firms advance along the compliance-efficiency-strategy trajectory, CP becomes embedded within organisational routines, innovation systems, and stakeholder engagement processes. By linking ecological responsibility with long-term competitive success, CP provides a viable pathway for firms seeking to thrive in an

increasingly sustainability-driven global economy.

5.6 Theoretical Contribution to the Natural Resource-Based View

This study extends the NRBV by suggesting that CP should be interpreted not merely as a technical environmental practice, but as an evolving capability with strategic implications. The findings indicate that the value of CP increases as it becomes embedded more deeply in organisational routines, governance arrangements, and innovation-oriented processes.

The contribution is twofold. First, the study introduces a stronger process perspective into the NRBV by showing that environmental capabilities develop gradually rather than existing as static resources. Second, it demonstrates that the strategic relevance of such capabilities is shaped not only by internal organisational factors but also by institutional conditions. In this sense, the study helps clarify how environmental practices may become part of longer-term strategic advantage.

6. Conclusion

This study examined how CP may evolve from a compliance-oriented practice into a strategic capability. Drawing on a multi-case analysis of six firms—Enerjisa, Tüpraş, Şişecam, Enel, E.ON, and TotalEnergies—the findings suggest a three-stage progression from compliance-driven adoption to efficiency-driven optimisation and, in more advanced cases, to strategy-driven integration.

As CP practices become more deeply embedded, they appear to generate both internal and external value. Internally, firms report process efficiencies, operational learning, and resource-related gains. Externally, they strengthen ESG-related positioning, stakeholder legitimacy, and broader sustainability alignment. In this sense, the findings suggest that CP may contribute to longer-term competitiveness when integrated into strategic decision-making and innovation processes.

The study contributes to the literature by interpreting CP as an evolving capability rather than a static environmental practice, linking it to qualitative signals of sustainable competitiveness, and providing comparative evidence broadly consistent with the Porter Hypothesis across different institutional settings. From a managerial perspective, the results suggest that firms are more likely to realise strategic value from CP when it is aligned with digital transformation, innovation activities, supply chain governance, and sustainability-oriented learning. For policymakers, the findings highlight the importance of regulatory coherence, consistent enforcement, green-finance incentives, and supportive disclosure frameworks in enabling firms to move beyond compliance and toward deeper strategic integration.

7. Limitations and Future Research

Despite its theoretical and empirical contributions, this study has several limitations. First, it adopts a qualitative multiple-case design based on six large, publicly listed, energy-intensive firms. While this approach allows for in-depth analysis and analytical generalisation, the findings are not intended to be statistically representative of all firms or sectors.

Second, the exclusive focus on large firms limits the transferability of the findings to SMEs, which often face different resource constraints, reporting capacities, and regulatory pressures. For this reason, the compliance–efficiency–strategy trajectory identified here may unfold differently in SME settings or in less energy-intensive industries.

Third, the analysis relies on secondary document-based evidence drawn from publicly available sustainability, integrated, and ESG reports. Although the study uses triangulation with third-party ESG assessments to reduce reporting bias, corporate disclosures may still contain symbolic or impression-management elements. Therefore, greenwashing risk cannot be fully excluded.

A further limitation concerns causal interpretation. Because the study is based on qualitative and document-based evidence over a limited time horizon, it does not establish direct causality between CP maturity and sustainable competitiveness outcomes. Instead, it identifies patterned associations across cases and institutional contexts.

Future research could address these limitations by using mixed-methods or quantitative designs, incorporating SME samples, extending the temporal scope, and integrating primary data such as interviews, survey evidence, or internal performance indicators. Comparative studies across sectors and regions would also help clarify how institutional conditions shape CP trajectories and their strategic implications.

Data Availability

The data that support the findings of this study are derived from publicly available reports, academic literature,

and organizational case studies, all cited throughout the article. Due to the qualitative and secondary nature of the analysis, no proprietary or raw datasets were generated or analyzed during the current study. All relevant sources are referenced in the manuscript and are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflicts of interest.

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Appendix

Appendix A. Coding Structure and Illustrative Examples

This appendix presents illustrative examples of the qualitative coding process used in the NVivo-based content analysis. The coding scheme was developed iteratively through open, axial, and selective coding, allowing analytical categories to emerge from corporate disclosures while remaining theoretically informed by the NRBV, dynamic capabilities logic, and the Porter Hypothesis.

Table A1 provides representative excerpts from corporate sustainability, integrated, and annual reports, together with the corresponding first-order codes, second-order categories, and CP integration stages. The purpose of this appendix is not to treat isolated excerpts as self-sufficient evidence, but to illustrate how repeated disclosure

patterns were translated into analytically comparable categories across cases and reporting periods.

In the coding process, stage assignment was based on theory-informed criteria rather than on single statements. Compliance-driven integration was identified when CP-related disclosures primarily emphasised legal adherence, certification, environmental management systems, and regulatory legitimacy. Efficiency-driven integration was identified when disclosures linked CP to resource optimisation, energy efficiency, recycling, digital monitoring, or process improvement. Strategy-driven integration was identified only when CP-related practices were explicitly connected to innovation, capital allocation, governance structures, long-term value creation, or strategic repositioning.

To strengthen interpretive transparency, the coding process also considered dynamic capability markers. In particular, disclosures were examined for evidence of sensing (recognition of sustainability-related pressures or opportunities), seizing (investment, redesign, or implementation responses), and reconfiguring (embedding CP into broader organisational, governance, or innovation systems). These indicators were used as interpretive supports rather than as mechanical classification tools.

Table A1. Illustrative coding examples from corporate disclosures

Sample Excerpt (Corporate Reports)	First-Order Code	Second-Order Category	CP Integration Stage
“Environmental compliance and adherence to national and international regulations remain the foundation of our sustainability approach, supported by ISO 14001-certified systems.” (Tüpraş, Integrated Report)	Regulatory compliance	Compliance-oriented CP	Compliance-driven integration
“Cullet recycling and alternative fuel use have reduced energy intensity and carbon emissions across selected production lines.” (Şişecam, Sustainability Report)	Energy efficiency improvement	Operational eco-efficiency	Efficiency-driven integration
“Digital metering and smart-grid investments enable real-time optimisation and systematic loss reduction.” (Enerjisa, Annual Report)	Digital optimisation	Process integration	Efficiency-driven integration
“Distribution networks are being transformed into intelligent platforms that manage complex energy and data flows, enabling efficient integration of renewable energy.” (E.ON, Integrated Annual Report)	Digital network transformation	System-level efficiency and integration	Advanced efficiency-driven integration
“Circular economy principles are embedded in infrastructure planning, product design, and capital allocation decisions.” (Enel, Integrated Annual Report)	Strategic embedding of CP	Strategic integration	Strategy-driven integration
“Efficiency gains from Cleaner Production (CP) initiatives are reinvested in low-carbon innovation pathways.” (TotalEnergies, Universal Registration Document)	CP-enabled reinvestment	Capability reconfiguration	Strategy-driven integration

The examples in Table A1 illustrate how CP can be interpreted as progressing from rule-based compliance mechanisms toward more deeply embedded organisational capabilities. Early-stage codes capture regulatory alignment and certification, whereas later-stage codes reflect operational learning, digital integration, strategic reinvestment, and stronger links to innovation and governance.

Importantly, these excerpts are presented as illustrative examples rather than exhaustive proof of stage membership. Final case classification was based on repeated patterns across multiple disclosures and reporting periods, as explained in the methodology and findings sections. In this respect, E.ON represents a boundary case: it exhibits advanced efficiency-driven integration at the infrastructure and system level, but the disclosures provide less explicit evidence of fully articulated corporate-wide strategic embedding than in Enel or TotalEnergies.

To enhance reliability, all documents were independently coded by two researchers. Inter-coder agreement was assessed using Cohen’s Kappa, yielding a value of 0.87, which indicates a high level of coding consistency. Discrepancies were resolved through joint discussion and codebook refinement.